

INCIDENT REPORT - 2

=====

Report ID: IR-002
Incident Name: Suspicious PowerShell Execution
Date/Time: May 4, 2026, around 21:21–21:28
Severity: HIGH
Status: Resolved (lab environment)
Environment: Wazuh 4.7.5 Mini SOC Home Lab
Analyst: Jazz

1. EXECUTIVE SUMMARY

At approximately May 4, 2026 around 21:21–21:28, Wazuh SIEM detected suspicious PowerShell execution on the Windows endpoint DESKTOP-HTP16PA. The alert showed PowerShell process activity, including encoded command execution and PowerShell spawning another PowerShell process.

PowerShell is a legitimate Windows administration tool, but attackers frequently abuse it for command execution, payload download, evasion, reconnaissance, and post-exploitation activity. Encoded PowerShell commands are especially suspicious because they hide the actual command content from quick inspection.

This activity was intentionally generated in the lab to test Wazuh's ability to detect suspicious Windows/Sysmon telemetry. The incident was confirmed as a true positive lab simulation.

The incident maps to MITRE ATT&CK T1059.001 — Command and Scripting Interpreter: PowerShell.

2. TIMELINE

May 4, 2026 around 21:20
Suspicious PowerShell test activity was generated on the Windows endpoint.

May 4, 2026 around 21:21
Wazuh detected PowerShell process behavior on DESKTOP-HTP16PA.

May 4, 2026 around 21:21
Wazuh generated a PowerShell process alert.

May 4, 2026 around 21:21–21:28

Wazuh generated alerts related to encoded PowerShell execution and suspicious process behavior.

May 4, 2026 after alert review

The alert was reviewed in Wazuh Security Events.

May 4, 2026 after investigation

The event was confirmed as a true positive lab simulation.

3. DETECTION SOURCE

Tool: Wazuh SIEM/XDR
Wazuh Version: 4.7.5
Detection Area: Windows suspicious command execution
Log Source: Windows Sysmon / Windows Event Logs
Affected Agent: DESKTOP-HTP16PA
Agent OS: Windows 10 Home
Agent IP: 192.168.56.101
Process: powershell.exe

Observed Wazuh Rule IDs:

- 92057 — PowerShell encoded command
- 92027 — PowerShell process spawned PowerShell instance

Severity:

- Rule 92057: Level 12
- Rule 92027: Level 4

Related Event:

- Sysmon Event ID 1 — Process creation

4. MITRE ATT&CK MAPPING

Tactic: Execution / Defense Evasion
Technique: Command and Scripting Interpreter
Sub-technique: T1059.001 — PowerShell

Explanation:

The activity maps to PowerShell execution because PowerShell was used to run suspicious command content. Encoded PowerShell also supports Defense Evasion because the command is hidden in Base64 format and is not easily readable.

Potential attacker next steps if successful:

- Download a malicious payload
- Execute malware
- Disable security tools
- Dump credentials
- Create persistence
- Move laterally to another system

5. INVESTIGATION FINDINGS

Source:

Local PowerShell activity on the Windows endpoint.

Target:

DESKTOP-HTP16PA Windows endpoint.

Observed behavior:

- PowerShell process activity was detected.
- A PowerShell process spawned another PowerShell process.
- Encoded command behavior was observed.
- Wazuh generated a high-severity alert.
- The alert appeared in Wazuh Security Events.
- The event was mapped to MITRE ATT&CK PowerShell execution.

Evidence reviewed:

- Wazuh Security Events showing PowerShell alerts
- Rule ID 92057 PowerShell encoded command
- Rule ID 92027 PowerShell process behavior
- Windows endpoint telemetry from DESKTOP-HTP16PA
- MITRE ATT&CK dashboard showing PowerShell-related activity

Compromise confirmed:

No. The command was intentionally generated in the lab.

Unauthorized execution confirmed:

No production unauthorized execution occurred.

6. RISK ASSESSMENT

Risk: HIGH

Reason:

Encoded PowerShell execution is high risk because attackers commonly use it for stealthy command execution, malware delivery, and post-exploitation activity. Even if PowerShell can be used legitimately, encoded commands should always be investigated.

The risk is especially high if:

- The command was run by a non-admin user
- The command downloaded content from the internet
- The parent process was unusual
- The command was followed by file creation
- The same pattern appears on multiple endpoints

In this lab, the risk was controlled because the activity was intentionally generated.

7. VERDICT

Verdict: TRUE POSITIVE — Lab Simulation

The alert is a true positive because the suspicious PowerShell behavior occurred and Wazuh detected it. This was not a false positive because the activity matched known suspicious PowerShell behavior.

In a real SOC, this alert should be treated as high priority until the command is decoded and confirmed safe or malicious.

8. RESPONSE ACTIONS TAKEN

1. Reviewed Wazuh Security Events.
2. Identified the PowerShell alert.
3. Verified the affected agent as DESKTOP-HTP16PA.
4. Verified Rule ID 92057 and severity Level 12.
5. Reviewed related PowerShell process activity.
6. Confirmed MITRE mapping to T1059.001.
7. Confirmed the event was lab-generated.
8. Documented the incident for SOC reporting practice.

9. RECOMMENDATIONS

1. Decode the Base64 PowerShell command.
2. Review Sysmon Event ID 1 process creation logs.
3. Check the parent process that launched PowerShell.
4. Monitor PowerShell command-line flags such as:
 - EncodedCommand
 - enc
 - NoProfile
 - ExecutionPolicy Bypass
5. Enable PowerShell Script Block Logging.
6. Restrict PowerShell usage for non-administrative users.
7. Monitor network connections created by PowerShell.
8. Search for the same command across other endpoints.
9. Isolate the endpoint if the activity is unauthorized.
10. Collect forensic evidence if malware execution is suspected.

10. FINAL ANALYST NOTE

This incident demonstrates Wazuh's Windows detection capability using Sysmon and Windows event telemetry. Wazuh successfully detected suspicious PowerShell execution, assigned a high severity level, and mapped the activity to MITRE ATT&CK. This is one of the strongest Windows detections in the Mini SOC project.